

Methodology Paper

External LLM Scoring Protocol for the Relevance-Gated Truth, Perceived-Truth, and Importance Framework

Version 3.5 - Methodology Only - No Results Included

Item	Specification
Framework evaluated	Relevance-gated truth-probability, perceived-truth, and importance scoring framework
Scoring model	GPT-5.5 Thinking
Model role	Blind external scorer operating as a highly unbiased, fairly skeptical expert using available evidence
Datasets covered	HumAID, CredEvent/TurkRatings, LIAR/PolitiFact
Primary outputs	T_abs(s), T_perc(s,r), I(s,r)
Document scope	Methodological structure, scoring protocol, leakage controls, research protocol, formula implementation, and reproducibility design
Excluded by design	Numerical results, dataset performance values, target-joined diagnostics, and post-hoc outcome tables

Methodological principle

The study does not treat the model as all-knowing. The scorer is asked to approximate the judgment of a highly unbiased, fairly skeptical expert constrained by the evidence available to it. This distinction is essential for T_abs: the output is evidence-weighted posterior probability under a defined epistemic stance, not metaphysical certainty.

Abstract

This methodology paper describes the empirical scoring protocol used to operationalize a relevance-gated framework for evaluating statements along three axes: evidence-weighted truth-probability, perceived truth-probability, and receiver-dependent importance. The protocol uses GPT-5.5 Thinking as a blind external scoring model. The model is not used as an all-knowing oracle; it is instructed to behave as a highly unbiased and fairly skeptical expert, updating from available evidence, source information, statement properties, delivery context, situational relevance, receiver properties, and corroboration. The methodology separates scorer-facing inputs from owner-side targets, applies dataset-specific scoring instructions, requires extensive non-verdict research for LIAR truth scoring, and preserves logs and manifests for auditability. This document intentionally contains no empirical results.

Term	Meaning in this methodology
T_abs(s)	Evidence-weighted truth-probability for statement s, evaluated by a highly unbiased, fairly skeptical expert using available evidence.
T_perc(s,r)	Perceived truth-probability of statement s for receiver r, anchored in T_abs but modified by delivery, situational context, and receiver properties.
I(s,r)	Receiver-dependent importance of statement s for receiver r; false statements may still be important when consequential.
Rel(s,c)	Relevance gate that prevents ambient high-stakes context from inflating importance unless the statement itself bears on those stakes.
Blind scoring	External scoring in which targets, labels, answer keys, and owner-side evaluation files are not available to the scorer.

1. Study Design Overview

The study is designed as a target-free external scoring exercise. Each row contains a statement or event item and a set of blind contextual fields. The scorer assigns atomic variables defined by the framework. The owner-side formula engine then transforms the atomic variables into the appropriate framework output. Target labels or human scores are held out until after scoring is complete.

- HumAID is used to operationalize receiver-dependent importance, $I(s,r)$, for a humanitarian crisis-monitoring receiver.
- CredEvent/TurkRatings is used to operationalize perceived truth-probability, $T_perc(s,r)$, for event-credibility judgments.
- LIAR/PolitiFact is used to operationalize evidence-weighted truth-probability, $T_abs(s)$, for political claims using external non-verdict research.
- All scorer-facing packets are target-free; owner-side evaluation is performed only after outputs and logs are returned.

2. Framework Structure

The underlying framework is organized around three outputs and six input clusters. The outputs form a nested architecture: T_abs uses source, statement, and corroboration evidence; T_perc adds delivery, situation, and receiver perception; I uses the full receiver-dependent consequence space and treats truth as only one dampened contributor.

Output	Primary question	Input scope	Dataset link
T_abs(s)	What probability would a highly unbiased, fairly skeptical expert assign to the statement given available evidence?	Source credibility, statement properties, corroboration, established knowledge, empirical evidence.	LIAR/PolitiFact
T_perc(s,r)	How true or credible does the statement appear to a receiver in context?	T_abs plus delivery context, situational context, receiver properties.	CredEvent/TurkRatings

I(s,r)	How consequential or attention-worthy is the statement for a receiver?	Delivery, relevance-gated situation, receiver consequence, and a dampened truth update.	HumAID
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Cluster	Variables	Function in scoring
1. Source credibility	Bias, expertise, rank, stated confidence	Determines how strongly source information updates truth probability.
2. Statement properties	Novelty, consistency, falsifiability, specificity	Determines whether the claim is coherent, testable, and bounded enough to update evidence.
3. Delivery context	Register, tone, posture/expression, timing, location, public/private setting	Modulates perceived truth and importance through how the statement is delivered.
4. Situational context	General situation, speaker stakes, audience, opposition, broad stakes, Rel(s,c)	Links surrounding stakes to the statement only when statement-specific relevance is present.
5. Receiver properties	Novelty to receiver, prior knowledge, relationship, cognitive state, personal/role consequence	Makes T_perc and I receiver-dependent.
6. Corroboration	Independent sources, established knowledge, empirical evidence	Provides the strongest direct evidence path for T_abs.

3. Model Used and Role Assignment

The scoring model specified for the external scoring protocol is GPT-5.5 Thinking. The model is used as a structured blind rater, not as a source of target labels. It is instructed to assign continuous atomic variable values according to the framework rubric and to provide row-level audit logs when research is required.

Aspect	Methodological specification
Model name	GPT-5.5 Thinking
Role	Blind scorer / evidence assessor / structured variable rater
Epistemic stance	Highly unbiased and fairly skeptical expert using available evidence
Not assumed	All-knowing oracle, benchmark-label memorizer, or fact-check verdict copier
Permitted output	Scorer-style CSV files and audit logs according to the packet schema
Disallowed output	Narrative-only summary, invented labels, use of target files, use of fact-check verdict pages

Why the model is described this way

The methodology distinguishes between evidence-weighted judgment and omniscience. A skeptical expert can update strongly when direct, independent, specific evidence is present, but must also record uncertainty when exact evidence is unavailable. This supports rigorous T_abs scoring without pretending the model has access to hidden ground truth.

4. Data Inputs and Dataset Mapping

The scorer-facing data are organized into dataset-specific blind inputs. The same framework variables are used across datasets, but the intended output differs by dataset. Dataset-specific instructions prevent the scorer from applying the wrong interpretation of variables such as personal consequence, social-media delivery, or evidential support.

Dataset	Scored construct	Receiver or epistemic target	Key methodological emphasis
HumAID	I(s,r)	Humanitarian or crisis-monitoring receiver	Role-based consequence, operational humanitarian relevance, distinction between disaster keywords and actionable relevance.

CredEvent/TurkRatings	T_perc(s,r)	General event-credibility receiver / crowd credibility judgment	Perceived credibility of event claims, public evidence cues, delivery/context, corroboration signals.
LIAR/PolitiFact	T_abs(s)	Highly unbiased skeptical expert	Exact-claim evidence resolution through extensive outside research, excluding fact-check verdict sources.

5. Scorer-Facing Packet Structure

The final recommended scorer packet is structured to be self-contained but target-free. It contains instructions, blind inputs, calibration cards, output templates, research protocols, and anti-leakage requirements. Owner-side targets, evaluation scripts with targets, and joined diagnostics are excluded from the scorer packet.

Packet component	Purpose
README / prompt-to-paste	Defines the task, output requirements, no-results/no-target rules, and files to return.
Protocol files	Define variables, scoring scale, calibration principles, and dataset-specific interpretations.
Blind inputs	Provide row-level content and context without target labels.
Calibration cards	Guide use of the full 0-1 scale without exposing empirical targets.
Output templates	Force scorer-style CSV outputs rather than free narrative summaries.
Research logs	For LIAR, require source-by-source and row-by-row audit trails.
Manifests/checksums	Support reproducibility and file integrity verification.

6. General Scoring Protocol

The external scorer follows a structured sequence for each row:

1. Read the statement, available context, source fields, and dataset-specific instructions.
2. Identify the intended output for the dataset: T_abs, T_perc, or I.
3. Assign atomic variables continuously on [0,1], using evidence-sensitive values rather than coarse categories.
4. Use the full scale when justified; avoid excessive shrinkage toward 0.5 when evidence is direct and strong.
5. Record uncertainty separately instead of using uncertainty as an automatic reason to collapse scores to the middle.
6. Flag rows only when evidence is genuinely insufficient or when leakage risk is detected.
7. Return only the required CSV/log files, not a narrative-only summary.

Anti-overconservatism rule

The scorer is instructed to avoid excessive epistemic conservatism or unwarranted posterior shrinkage. In scientific terms, direct, independent, specific, and high-quality evidence should produce strong updates. Residual uncertainty should be documented in logs, not used mechanically to force midrange scores.

7. Dataset-Specific Protocols

7.1 HumAID: Importance Scoring

HumAID rows are scored for I(s,r), where the receiver is a humanitarian or crisis-monitoring role. The key methodological point is that AC is role-based consequence, not only private personal danger to a random reader. A post may be highly important if it indicates urgent needs, injury, death, displacement, infrastructure disruption, missing persons, or other decision-relevant crisis information.

- High importance requires statement-specific relevance, not mere co-location with a disaster event.
- Disaster keywords alone are insufficient; the statement must carry operational, humanitarian, evidential, or decision-relevant content.
- Public social-media posts should not automatically receive very low X or W merely because the platform is informal.
- Unavailable nonverbal cues should usually be treated as neutral or minimally informative, not as strong negative evidence.

7.2 CredEvent/TurkRatings: Perceived Truth Scoring

CredEvent/TurkRatings rows are scored for $T_{perc}(s,r)$. The task is to estimate how credible an event claim appears to receivers given the available public context, source cues, corroboration signals, and qualitative reasoning fields. The scorer should distinguish between evidence that supports the event and evidence that merely makes the event plausible.

- The publicness of event-stream claims should be reflected in X and delivery/context variables.
- Corroboration and consistency should be rewarded when independent or reputable evidence is present.
- Opposition or contradiction should reduce perceived truth when it is active and evidence-based.

7.3 LIAR/PolitiFact: Evidence-Weighted Truth Scoring

LIAR rows are scored for $T_{abs}(s)$. Unlike HumAID and CredEvent, broad context is not sufficient. The scorer must resolve the exact claim using outside research. The evidence search is mandatory and extensive, but must avoid fact-check verdict sources and any benchmark labels.

Allowed source types	Prohibited source types
Official statistics, government datasets, agency records, legislation, roll-call votes, bill histories, transcripts, archived pages, court records, academic/nonpartisan research, reputable normal news when primary records are unavailable.	PolitiFact verdict/rating pages, Truth-O-Meter pages, LIAR labels, Snopes, FactCheck.org, Washington Post Fact Checker, AP/Reuters/AFP fact-check verdict pages, benchmark datasets, target files, prior scorer outputs, or any verdict/label source.

- Exact-claim support must be distinguished from general topic support.
- Partial truth should be scored as partial support, not forced into binary true/false form.
- Evidence logs must identify the sources searched, source types, and unresolved components.
- Rows that cannot be resolved after extensive search must be marked as unresolved rather than guessed.

8. Leakage Control and Blindness

The protocol separates scorer-side materials from owner-side evaluation materials. Scorer packets contain no target labels, no joined audit files, no performance metrics, and no owner-side diagnostics. Output files are evaluated only after scoring is complete.

Risk	Control
Target leakage	Targets, labels, and answer keys are excluded from scorer packets.
Fact-check verdict copying	Verdict pages and fact-check rating sources are explicitly banned for LIAR scoring.
Narrative-only response	Output templates require structured scorer-style CSV files and logs.
Overuse of broad context	LIAR protocol requires exact-claim resolution; HumAID protocol requires statement-to-stakes relevance.
Unlogged evidence use	Research logs and source usage logs are required for LIAR.
File drift	Versioned archives and checksum manifests are used.

9. Owner-Side Formula Implementation

After the external scorer returns atomic variables, the owner-side engine computes the framework outputs. The scorer does not directly set final dataset predictions except through the atomic variables it supplies. This preserves a separation between evidence assessment and mathematical aggregation.

```
T_abs(s) = BayesianUpdate(Prior, Cred, Stmt, K1..Kn, L, M)
T_perc(s,r) = BayesianUpdate(T_abs, Deliv, Sit*, Recv)
I(s,r) = BayesianUpdate(0.50, Deliv_tilde, Sit*_tilde, Recv_tilde, dampened T_abs update)
```

The implementation applies root normalization to multiplicative composites before likelihood-style or tilde transformations. This prevents mechanical compression caused by multiplying several [0,1] variables.

```
Deliv_final = sqrt(Deliv_raw)
Sit_final   = sqrt(Sit_raw)
Recv_final  = cbrt(Recv_raw)

Deliv_tilde = tilde(Deliv_final)
Sit_tilde   = tilde(Sit_final)
Recv_tilde  = tilde(Recv_final)
```

When no independent corroborating sources are present, the K update is skipped rather than interpreted as negative evidence. Raw novelty values are transformed internally into the framework-specific novelty terms for perceived truth and importance.

```
if K_count == 0:
    skip corroboration update
else:
    apply K update using K_mean and corroboration decay

N_perc = inverted-U transform of N_raw
N_imp  = monotonic importance transform of N_raw
```

10. Research Logging and Audit Requirements

For LIAR, each scored row requires a row-level research log and a source usage log. These logs are methodological artifacts: they allow the owner to audit whether the scorer used non-verdict evidence, whether the exact claim was resolved, and whether uncertainty was justified.

Required file	Purpose
liar_atomic_scores_raw_from_external_llm.csv	Main atomic variable scores used by the formula engine.
liar_row_research_log.csv	Row-by-row research summary, exact-claim decomposition, evidence direction, and unresolved components.
liar_source_usage_log.csv	Source-level log documenting domains, source types, and usage roles.
liar_unresolved_rows.csv	Rows that could not be adequately resolved after research.
scoring_session_log.md	Global session notes, limitations, compliance statements, and deviations.

11. Evaluation Plan

The owner-side evaluation compares computed framework outputs to held-out dataset targets only after scoring is complete. This methodology paper describes the evaluation metrics but intentionally reports no numerical results.

Metric	Purpose
Spearman correlation	Assesses monotonic ranking agreement between model outputs and targets.

Pearson correlation	Assesses linear association between model outputs and targets.
Mean Absolute Error (MAE)	Assesses absolute calibration error on the normalized target scale.
Graded AUC / concordance index	Assesses pairwise ranking performance when targets are continuous or ordinal.
Binary ROC-AUC	Assesses discrimination at predefined binary thresholds when thresholds are meaningful.
Group diagnostics	Assesses behavior across hidden target groups after evaluation, without exposing groups to the scorer.
Leakage/source audit	Checks whether scorer outputs or logs used prohibited target or verdict sources.

No results included

This methodology document names the metrics and audit procedures, but it deliberately omits all numerical performance values, group means, AUC values, error tables, and target-joined diagnostics.

12. Versioning and Protocol Refinement

The scoring protocol was refined through target-free packet revisions. The purpose of versioning is to document methodological improvements without changing the target labels or exposing them to scorers.

Version	Methodological role
V3 core	Introduced calibrated scorer packet for HumAID and CredEvent with target-free blind inputs and output templates.
V3.2 addendum	Added anti-keyword guidance for HumAID to reduce disaster-adjacent false positives and clarify statement-specific humanitarian relevance.
V3.3 LIAR addendum	Added broad non-verdict evidence leads for LIAR while keeping labels and fact-check verdicts excluded.
V3.4 LIAR research packet	Made outside research required and added row-level logs, source usage logs, and unresolved-row reporting.
V3.5 LIAR extensive research packet	Expanded research freedom while maintaining fact-check bans, added exact-claim resolution, and added anti-overconservatism calibration language.

Reason for LIAR refinement

Early LIAR scoring was methodologically too conservative because broad evidence alone did not resolve exact truth claims. The final LIAR protocol therefore requires extensive outside research, while still banning fact-check verdict pages and benchmark labels.

13. Reproducibility and File Governance

The workflow uses separate bundles for scorer inputs, external outputs/logs, owner-side analysis, and final methodology documentation. This separation makes it clear which files can be sent to external scorers and which files must remain private because they contain targets, joined diagnostics, or evaluation results.

Bundle type	Contents	May be sent to external scorer?
Scorer input bundle	Target-free prompts, blind inputs, calibration instructions, output templates, research protocols.	Yes
External outputs/logs archive	Raw files returned by external scorers and logs, preserved as received.	Usually no; archival/audit use.
Owner analysis bundle	Formula scripts, joined audit files, metrics code, and diagnostics.	No
Methodology paper	Protocol-only document; no results.	Yes

14. Methodological Limitations

- The LLM scorer can only update from evidence surfaced through the packet and its permitted research, so T_{abs} is evidence-weighted rather than all-knowing truth.
- Row-level truth scoring for LIAR depends on the quality and completeness of research logs and source access.
- The use of a single model family as scorer limits inter-rater reliability unless future studies add independent human or model raters.
- Dataset targets are proxies for the framework outputs; they are useful validation targets but not perfect measurements of the theoretical constructs.
- Some continuous variables remain rubric-scored and therefore require future calibration studies and inter-rater agreement analysis.

15. Conclusion

This methodology defines a leakage-controlled external LLM scoring protocol for a relevance-gated truth, perceived-truth, and importance framework. GPT-5.5 Thinking is used as a blind structured scorer under a skeptical-expert epistemic stance. The protocol maps HumAID to importance, CredEvent/TurkRatings to perceived truth, and LIAR/PolitiFact to evidence-weighted truth. It separates target-free scoring from owner-side evaluation, requires extensive non-verdict research for truth claims, documents all scorer outputs and logs, and preserves reproducibility through versioning and checksums. No empirical results are included in this paper.

Appendix A. Scorer Return Contract

The expected scorer return contract consists of structured files, not narrative summaries:

- Atomic scores CSV for each scored dataset
- Research logs for LIAR
- Source usage logs for LIAR
- Unresolved-row file for LIAR
- Scoring session log
- No target labels, no verdict pages, no owner-side diagnostics